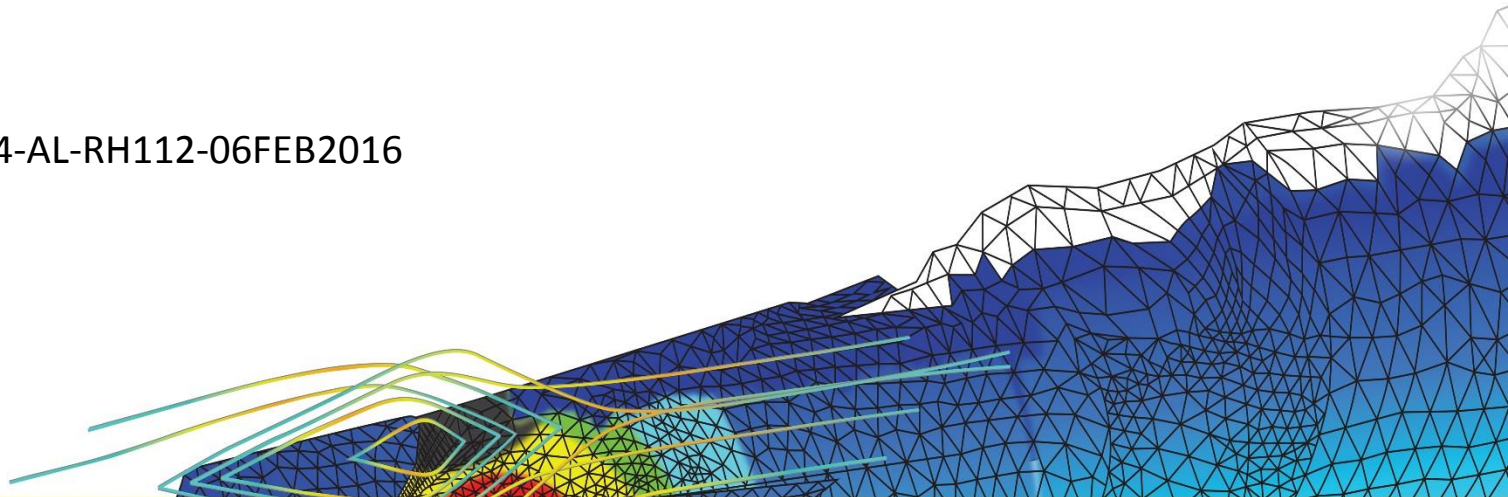




RedHawk™ Low Power Analysis

VERSION: V6.4-AL-RH112-06FEB2016

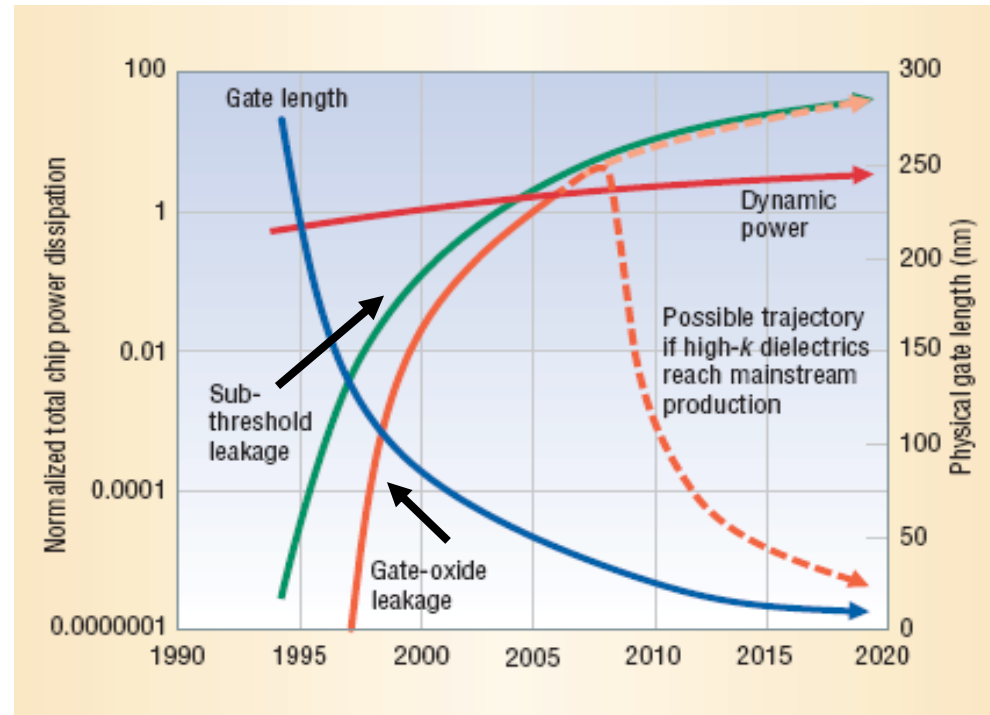


Agenda

- **Low power analysis theory**
- **RedHawk-LP Lab**
- **Summary**

Controlling Leakage Power

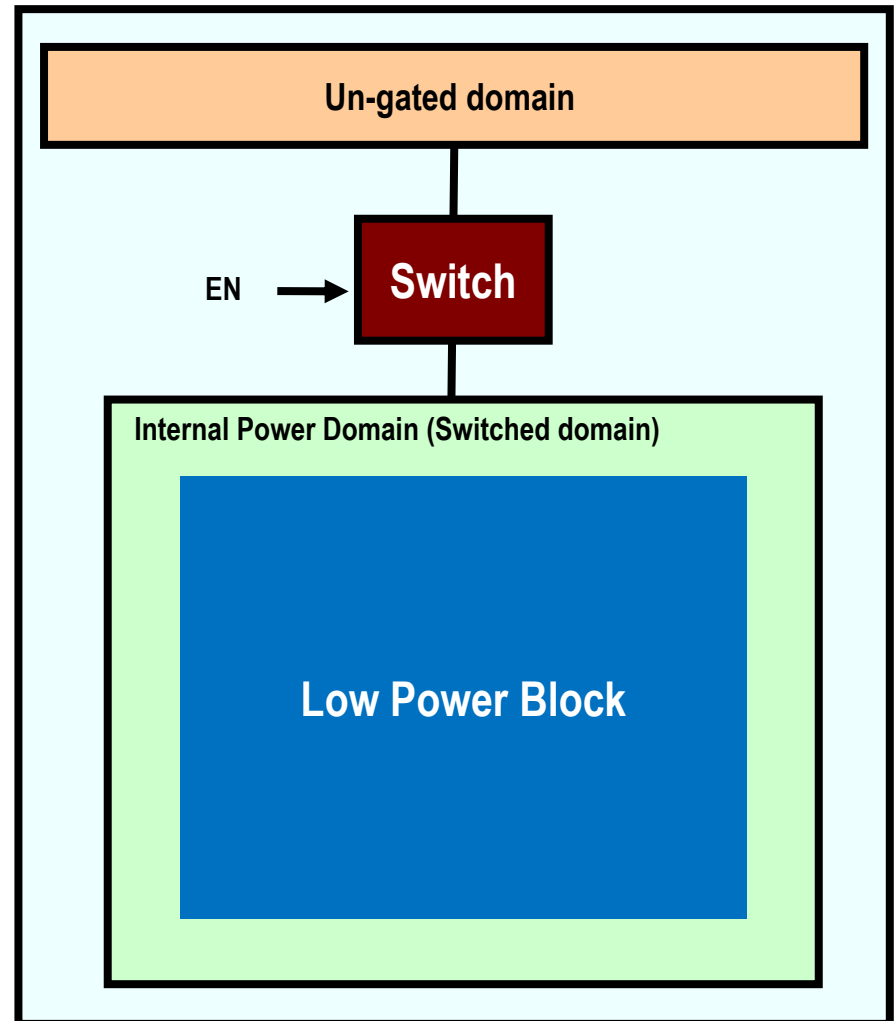
- > 50 % of total power goes to leakage
- **Cell phone battery life**
- Standby power cannot exceed 5% of full operational power
- **Product's competitiveness depends on its ability to control leakage**



Kim et al, "Leakage Current: Moore's Law Meets Static Power," *IEEE Transactions on Computers*, Vol. 36, No. 12, December 2003, pp. 68-77

Leakage Power Saving Using MTCMOS Switches

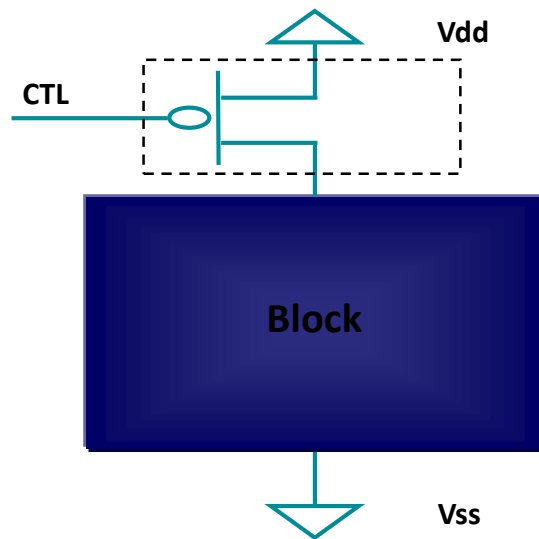
- Can turn off the power for inactive blocks using switches
- Switches are made with High V_t transistors for minimal sub-threshold leakage



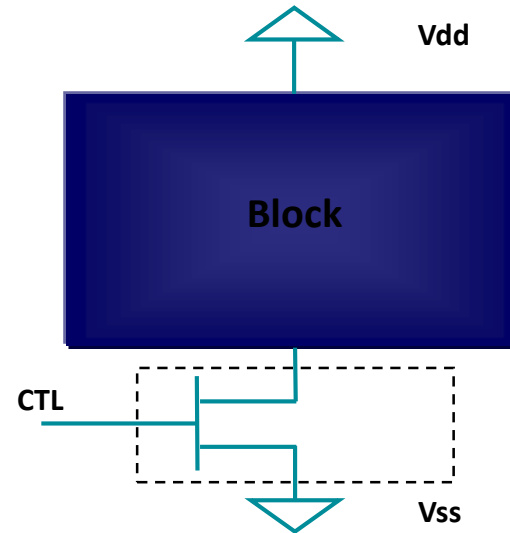
Different Switch Architectures

Power Gating Switch Models

Header Switches (PMOS)



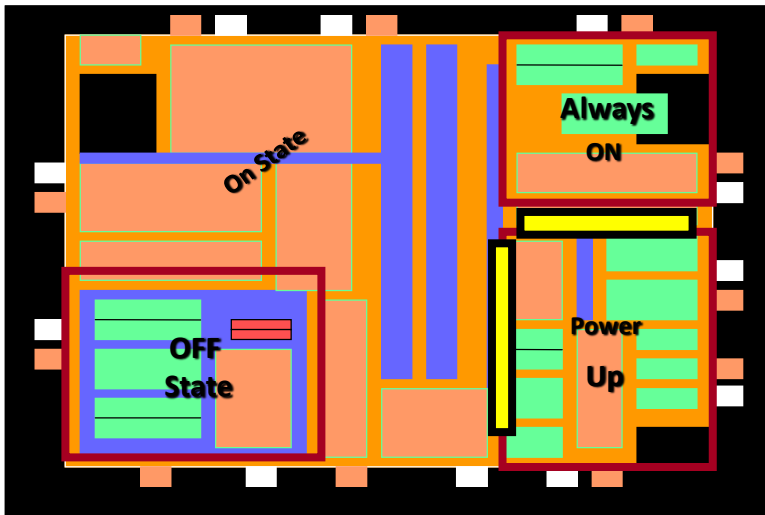
Footer Switches (NMOS)



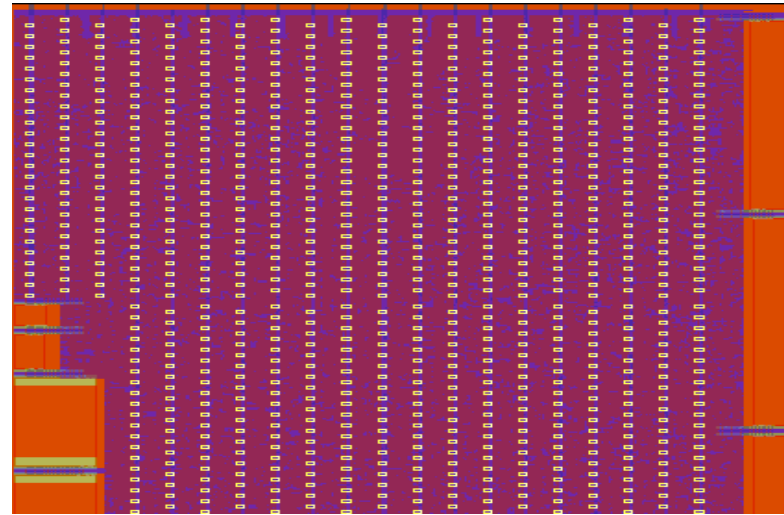
Different Switch Implementation Methods

Full-chip Implementation

Coarse Grain



Fine Grain



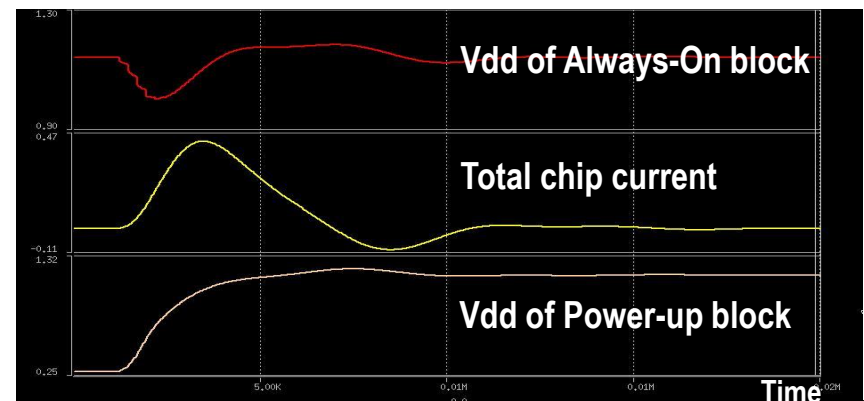
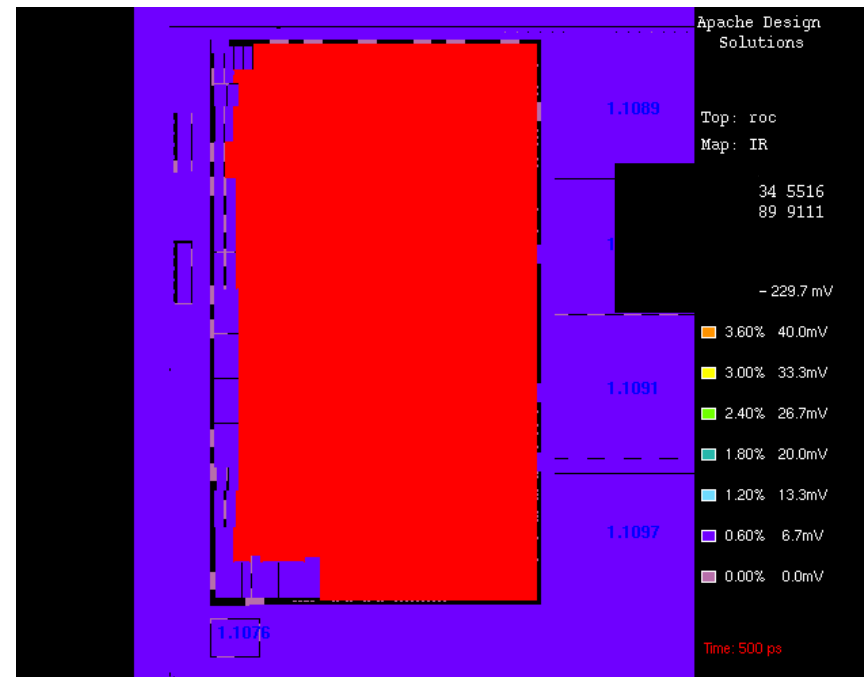
Switches highlighted in yellow

Different Switch Operating States

Mode	Comment
OFF	When block is inactive
ON	When block is functional
Powerup	When block is transitioning from inactive to functional state
Powerdown	When block is transitioning from functional state to inactive state

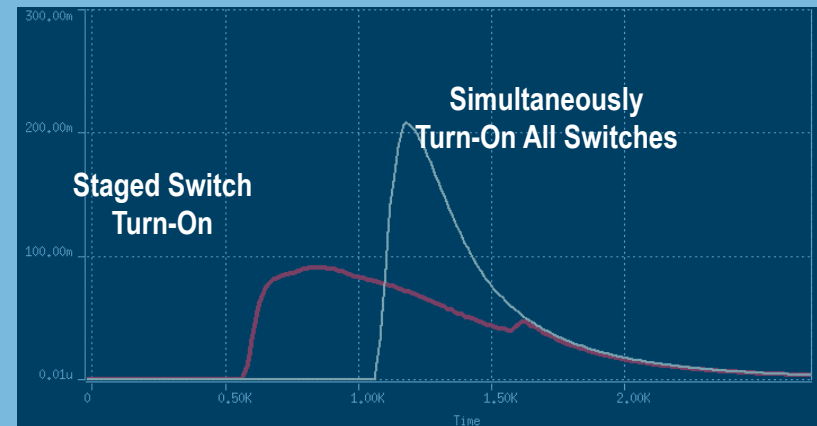
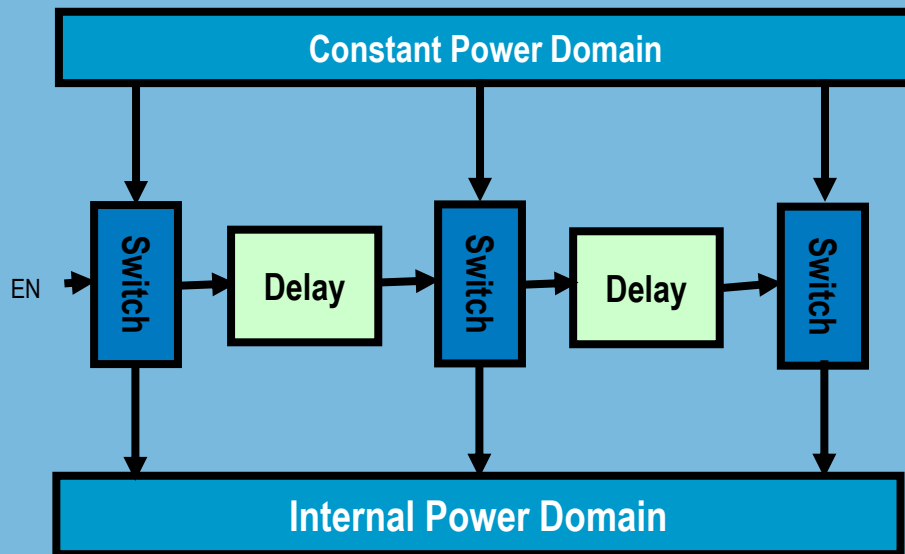
Challenges in Power Gated Designs

- High surge of current to charge/ discharge the internal capacitors during powerup
- This can cause huge voltage drop on neighboring “always on” blocks
- Peak current flowing through the switches should be within limits



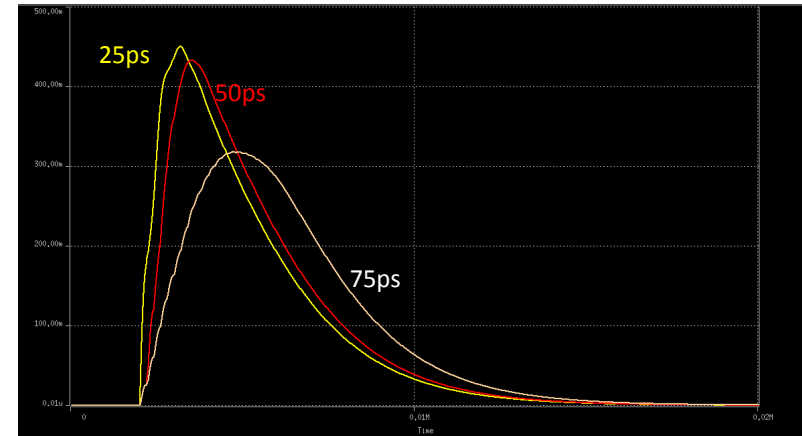
Challenges in Power Gated Designs (Cont'd)

- Daisy chain structure for switch enable helps reducing rush current

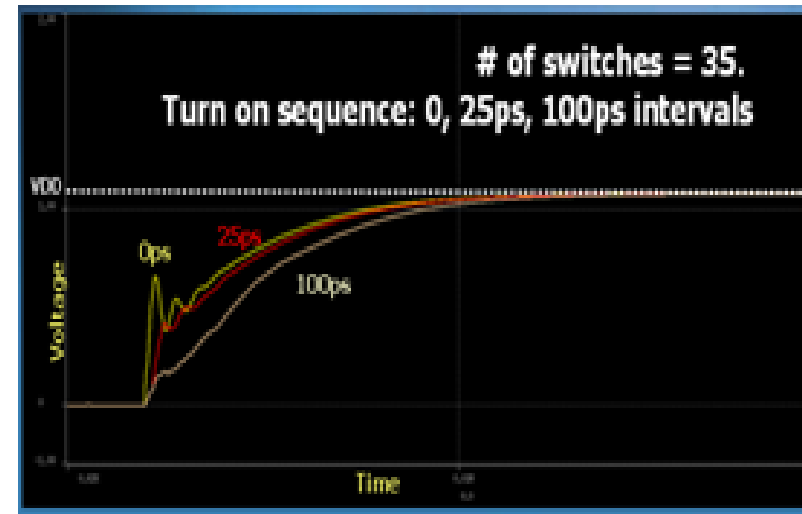


Challenges in Power Gated Designs (Cont'd)

- Rush current can be reduced by increasing the delay between switches

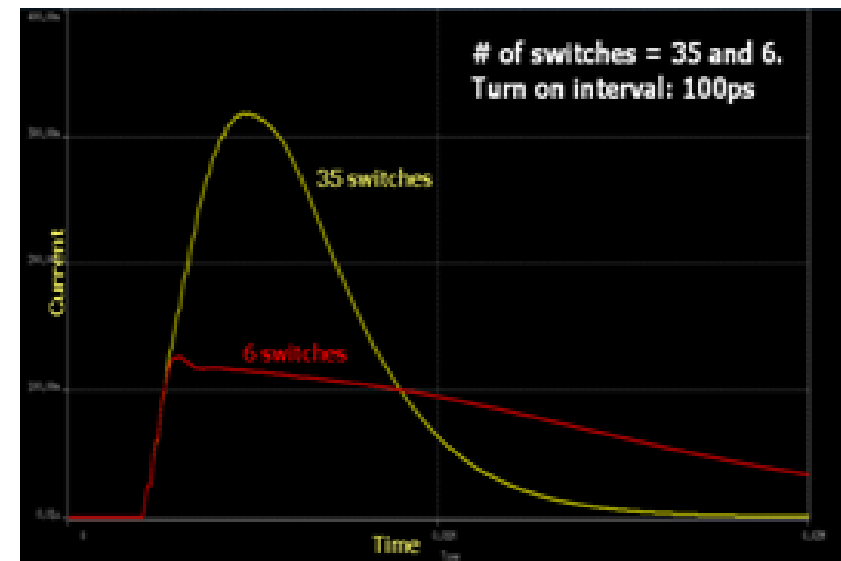


- Increasing the delay will also cause an increase in block wake-up time



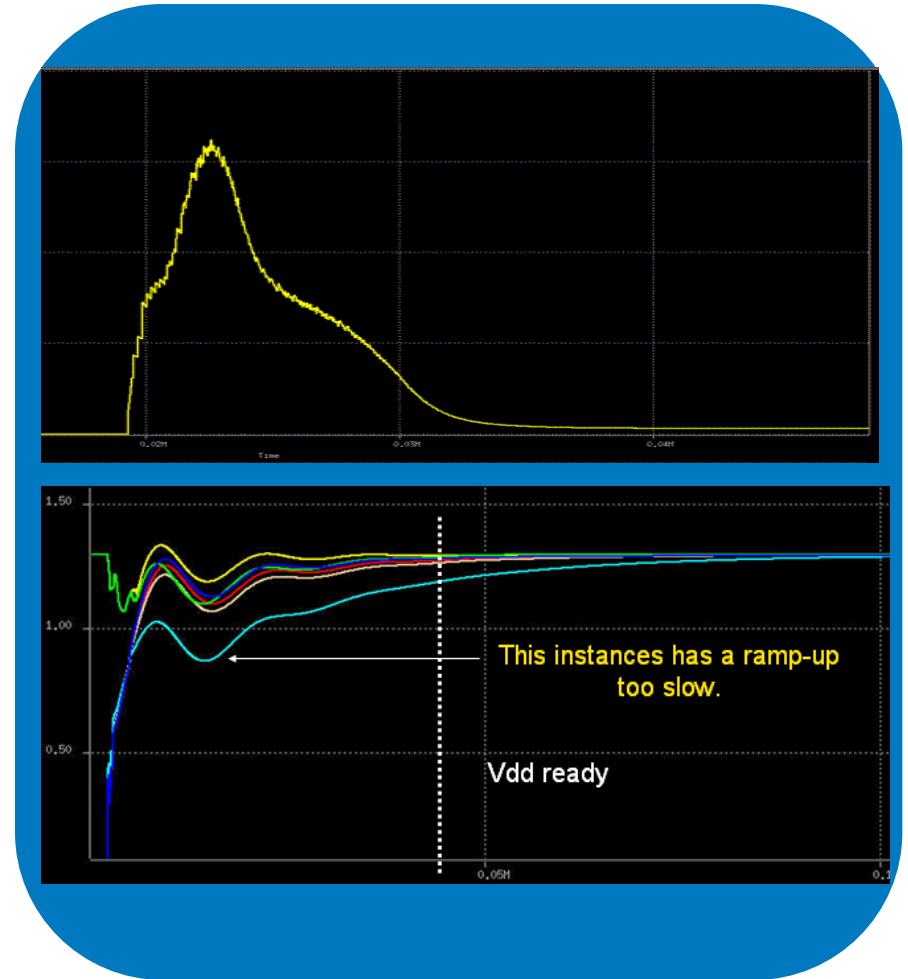
Challenges in Power Gated Designs (Cont'd)

- Rush current can be reduced by reducing the number of switches
- But this will increase the current flow through individual switches
- Also increases the ON state power grid resistance
- Increases the wake-up time



Challenges in Power Gated Designs (Cont'd)

- High rush current can cause oscillations in the power ground n/w due to fast turn ON ($L di/dt$)



Challenges in Power Gated Designs (*Cont'd*)

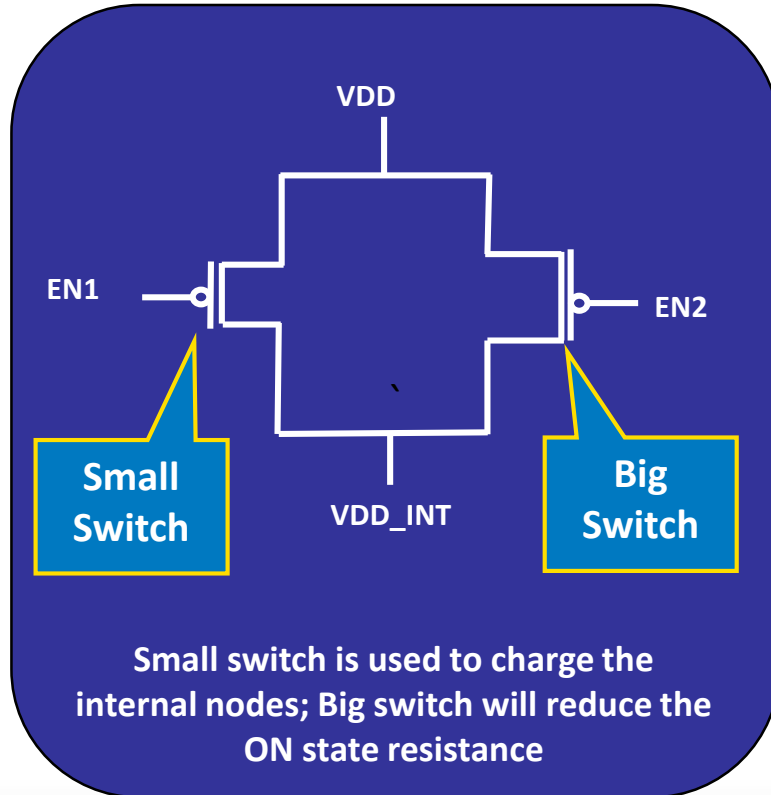
- Power switches add more resistance in the power grid
- Increase in static/dynamic voltage drop in ON state analysis
- Drop increases if you reduce the number of switches
- Hot spot regions will require more number of switches

Can we increase the number of switches to reduce the ON state drop ?

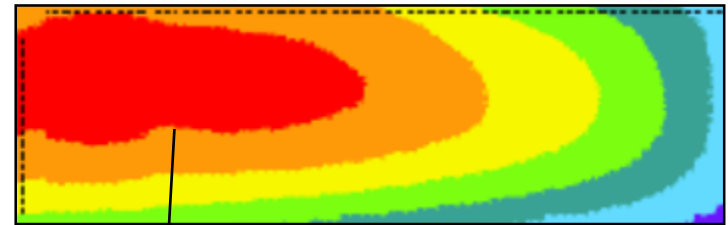
- Increase in area
- More off state leakage current
- High rush current

Challenges in Power Gated Designs (Cont'd)

- Alternate switch architecture to reduce rush current and ON state voltage drop



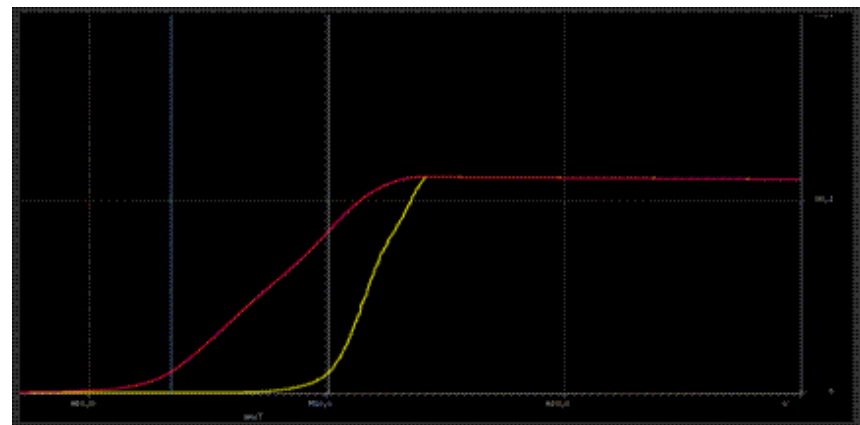
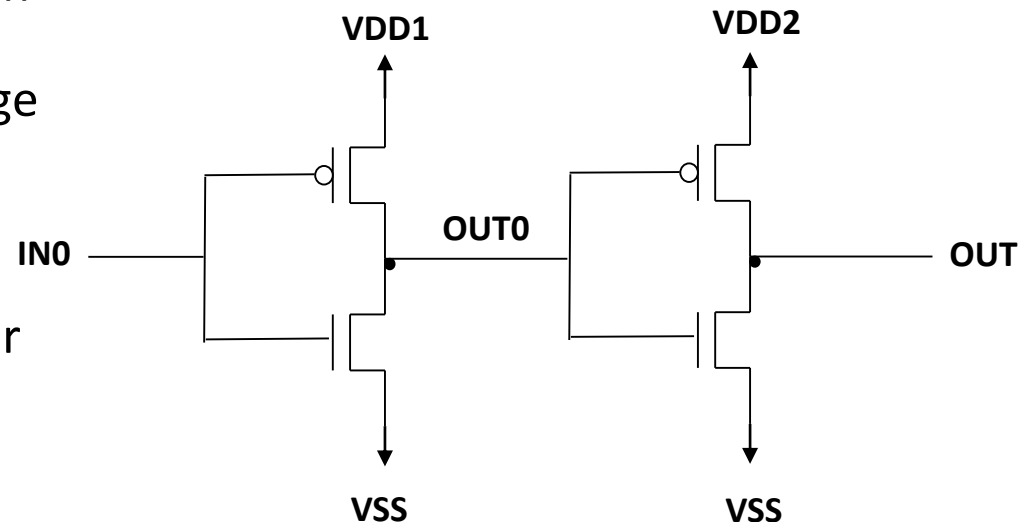
- Switch placement can be improved to optimize the ON state drop



More switches can be used in critical areas to reduce the drop

Challenges in Power Gated Designs (Cont'd)

- If VDD2 ramps up faster than VDD1, OUT0 can stay at a level above threshold voltage for more time.
- There could be high crowbar current in the receiver cell
- More significant when wakeup time is more
- Need to analyze the voltage differential analysis on driver receiver pair



RedHawk-LP Features

ON state Analysis

Static IR drop analysis

Dynamic analysis

OFF state Analysis

Leakage

Powerup Analysis

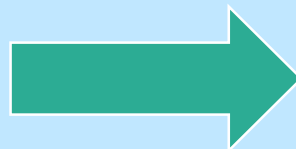
Rush current

Noise coupling

Wake-up time

Crowbar current

P/G n/w oscillations



Switch size and architecture

Number of switches

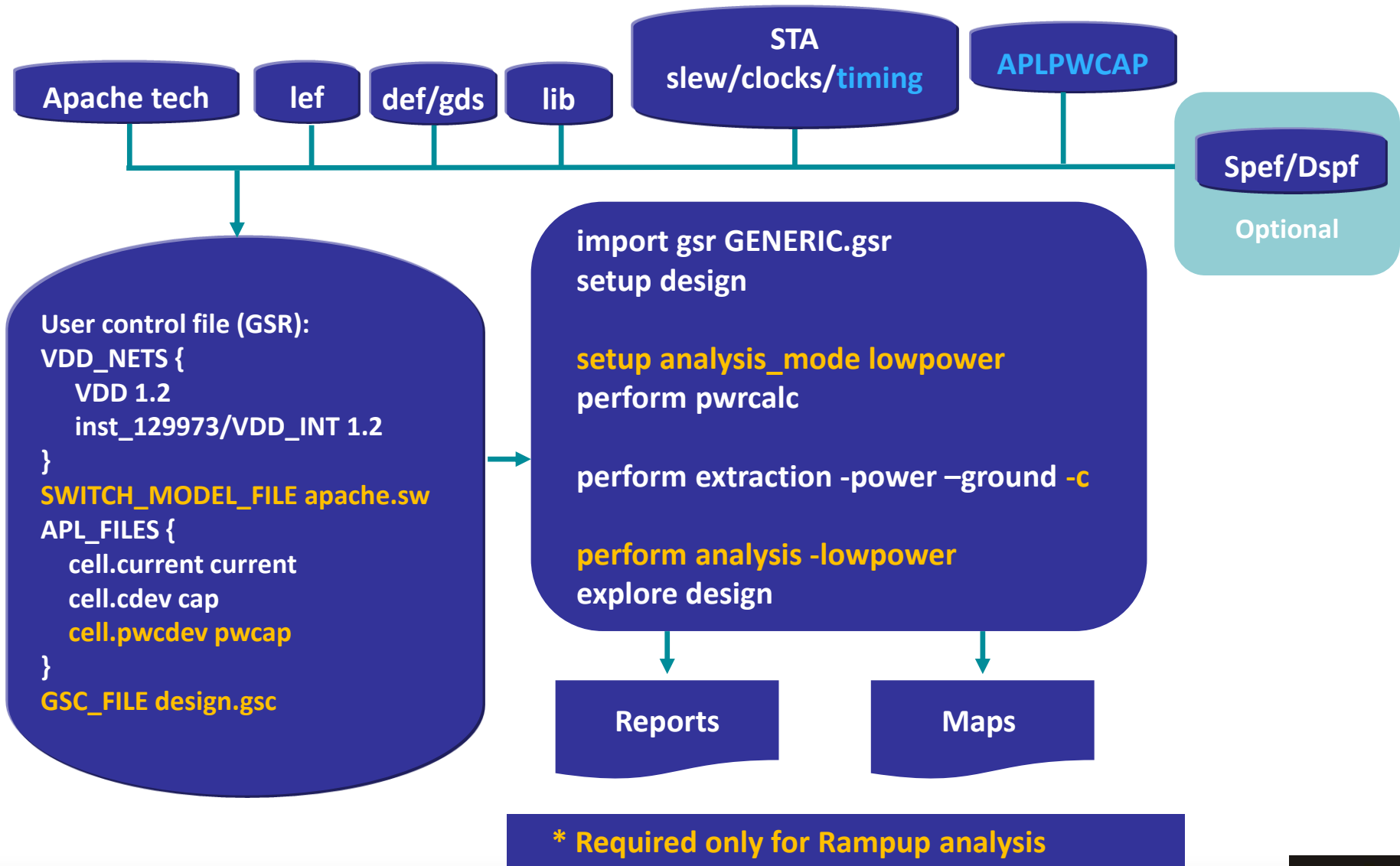
Placement of switches

Interval between switches

Switch turn ON sequence

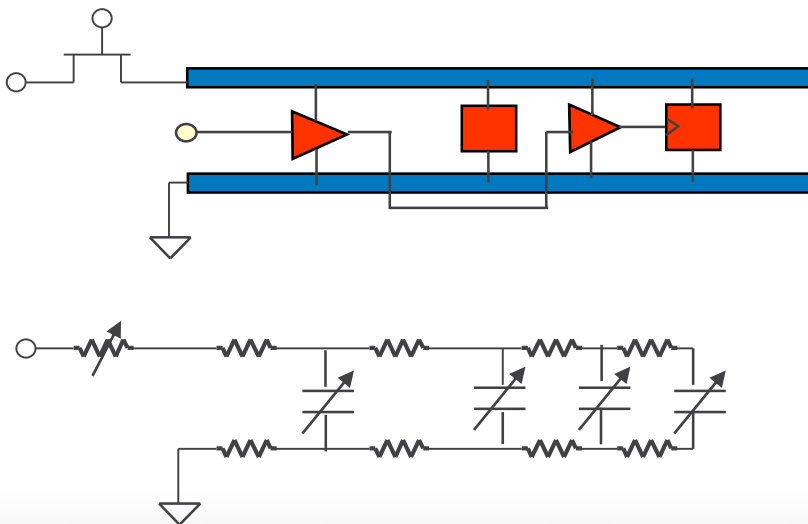
DATA REQUIREMENTS FOR LOW POWER ANALYSIS

RedHawk – LP



Additional Data Requirements: *APL PWCAP Data*

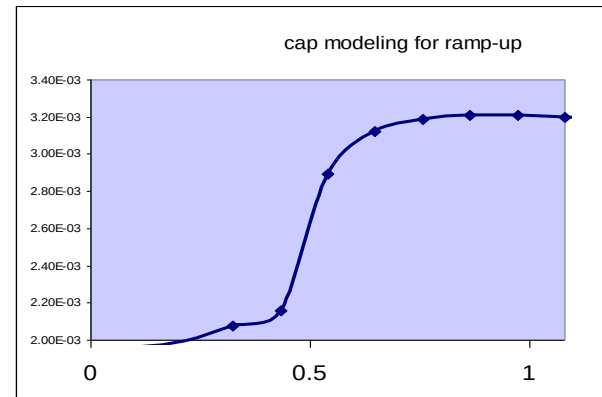
- Power gates are non-linear elements in different modes
- Transition from “off” to “on” state requires charge transfer to/from design
- Design capacitance varies as function of node voltages
- Long simulation times to capture entire power-up process
- Required only for Powerup analysis; Not required for ON state analysis



**Actual
Circuit**

**RedHawk
Equivalent
Circuit**

Non-linear capacitor modeling



Additional Data Requirements:

STA File with Switch Turn ON time

- STA File should contain turn ON time for the switch enable pins
- Required only for Rampup analysis; Not required for ON state analysis

```
switch_inst_R13_C1/CNT_IN1 TW 0 3.6e-9 3.6e-9 3.6e-9 3.6e-9 0
switch_inst_R14_C1/CNT_IN1 TW 0 3.8e-9 3.8e-9 3.8e-9 3.8e-9 0
switch_inst_R15_C1/CNT_IN1 TW 0 4e-9 4e-9 4e-9 4e-9 0
switch_inst_R16_C1/CNT_IN1 TW 0 4.2e-9 4.2e-9 4.2e-9 4.2e-9 0
switch_inst_R17_C1/CNT_IN1 TW 0 4.4e-9 4.4e-9 4.4e-9 4.4e-9 0
switch_inst_R1_C2/CNT_IN1 TW 0 4.6e-9 4.6e-9 4.6e-9 4.6e-9 0
switch_inst_R2_C2/CNT_IN1 TW 0 4.8e-9 4.8e-9 4.8e-9 4.8e-9 0
switch_inst_R3_C2/CNT_IN1 TW 0 5e-9 5e-9 5e-9 5e-9 0
switch_inst_R4_C2/CNT_IN1 TW 0 5.2e-9 5.2e-9 5.2e-9 5.2e-9 0
switch_inst_R5_C2/CNT_IN1 TW 0 5.4e-9 5.4e-9 5.4e-9 5.4e-9 0
switch_inst_R6_C2/CNT_IN1 TW 0 5.6e-9 5.6e-9 5.6e-9 5.6e-9 0
switch_inst_R7_C2/CNT_IN1 TW 0 5.8e-9 5.8e-9 5.8e-9 5.8e-9 0
switch_inst_R8_C2/CNT_IN1 TW 0 6e-9 6e-9 6e-9 6e-9 0
```

```
set ADS_CELLS_NEED_INPUT_TW { switch }
getSTA * -o design.timing
```

Additional Data Requirements:

Switch Model File

- Need to run APL characterization for switches to create the electrical model
- Captures the switch characteristics for different operating modes
- Required for both Rampup and ON state analysis

Review the switch model file at :
[design_data/apl/switch.model](#)

```
SWITCH_CELL switch {  
  SWITCH_TYPE: HEADER  
  EXT_PIN: VDD_EXT  
  INT_PIN: VDD_INT  
  CTRL_PIN: CNT_IN1 R -  
  
  ON:  
    R 20  
    I 3.226e-08  
    C 1.061e-13  
    IDSAT 0.01169  
  
  OFF:  
    C 3.174e-20  
    VI 1.2  
    N_VO_I 12  
    0 3.588e-09  
    ***  
    0.09167 3.348e-09  
    ***  
    0.1833 3.117e-09  
    ***  
    1.089 4.297e-08  
    ***  
    1.189 4.597e-08  
    ***  
    1.200 4.600e-08  
    ***  
  POWER_UP:  
    VI 1.2  
    N_VO_R 12  
    0 317.5  
    ***  
    0.09167 294.2  
    ***  
    0.1833 270.6  
    ***  
    1.089 61.36  
    ***  
    1.189 25.00  
    ***  
    1.2000 20.00  
    ***  
  POWER_DOWN:  
    VI 1.2  
    N_VO_R 12  
    0 104.4
```

Additional Data Requirements:

GSC File

- **Global switching constraint (GSC) file to specify the operating mode for the switch**
 - Specify the state of BOTH switch instance and power-gated block/instance:
 - [gated_block] POWERUP
 - [switch_instance] POWERUP
 - Or specify domain_name to include everything in that domain
 - * [domain_name] POWERUP
- **Can perform analysis for multiple scenarios using “import gsc” command**
- **GSC file is optional for ON state analysis. By default, all switches are assumed to be in ON state**

```
* POWERUP
switch_inst_R*_C1 POWERUP
switch_inst_R*_C2 OFF
switch_inst_R*_C3 POWERUP
switch_inst_R*_C4 OFF
switch_inst_R*_C5 POWERUP
switch_inst_R*_C6 OFF
switch_inst_R*_C7 POWERUP
switch_inst_R*_C8 OFF
switch_inst_R*_C9 POWERUP
switch_inst_R*_C10 OFF
switch_inst_R*_C11 POWERUP
switch_inst_R*_C12 OFF
switch_inst_R*_C13 POWERUP
switch_inst_R*_C14 OFF
switch_inst_R*_C15 POWERUP
switch_inst_R*_C16 OFF
switch_inst_R*_C17 POWERUP
switch_inst_R*_C18 OFF
switch_inst_R*_C19 POWERUP
switch_inst_R*_C20 OFF
switch_inst_R*_C21 POWERUP
switch_inst_R*_C22 OFF
```

Additional Data Requirements:

Command File / GSR Changes

- Increased simulation time and relaxed time step required for capturing the entire powerup simulation (Only for rampup)
- "setup analysis_mode lowpower" to optimize the extraction for low power analysis (Only for rampup)
- GSC file to be included in GSR or command file (only for rampup)

Additional Data Requirements:

Command File / GSR Changes (Cont'd)

- **Switch model file to be included in GSR (ON state and Rampup)**
- **STA file with switch timing file needs to be included in GSR or command file (Only for rampup)**
- **Virtual domain to be included in VDD_NETS/GND_NETS section (ON state and Rampup)**

ALP3D

- **ALP3D feature can be used to speed-up for ramp-up analysis**
 - ~3X overall speed-up in rampup simulation
- **Trade-off some accuracy loss**
 - QoR impact is within 10 – 15 %
- **ALP3D is invoked with the following command:**
 - `"perform analysis -lowpower -alp3d"`

REDHAWK-LP LAB

RedHawk – LP Lab

- **Mixed Mode Analysis**
 - cd low_power/mixed_mode
 - redhawk -f run_mixed_mode.tcl
- **Review the GSR and Command File**
 - Virtual domain included in GSR
 - Switch model file
 - GSC file
 - STA file with switch timings
 - setup analysis_mode lowpower

```
import gsr GENERIC.gsr
setup design
```

```
# Calculate power
perform pwrcalc
```

```
setup analysis_mode lowpower
```

```
# Power grid extraction
perform extraction -power -ground -c
```

```
# Lumped resistance(in Ohms) for package,wirebond
# and pads
```

```
# Default values are all 0, i.e no off-chip impact
```

```
setup package -power -r 0.005 -l 2.5 -c 5
```

```
setup package -ground -r 0.005 -l 2.5 -c 5
```

```
setup wirebond -power -r 0.01 -l 2.2 -c 1.42
```

```
-ground -r 0.05 -l 1.7 -c 0.2
```

```
setup pad -power -r 0.001
```

```
setup pad -ground -r 0.001
```

```
#
```

```
perform analysis -lowpower
```

```
explore design
```

Command File

GSR Review

```
SWITCH_MODEL_FILE ../design_data/apl/switch.model
GSC_FILE GENERIC.gsc
LEF_FILES {
  ../design_data/lef/switch.lef.gz
STA_FILE {
  GENERIC ../design_data/timing/GENERIC.timing.gz
APL_FILES {
  ../design_data/apl/stdcells_aplpwcap/cell.pwcap pwcap
  ../design_data/apl/stdcells_aplpwcap/decap_cell.pwcap pwcap
}
DYNAMIC_SIMULATION TIME 15e-9
DYNAMIC_TIME_STEP 50e-12
```

Switch Model

GSC File

Switch Lef File

STA file with switch timings

APL Pwcap files

More Simulation time

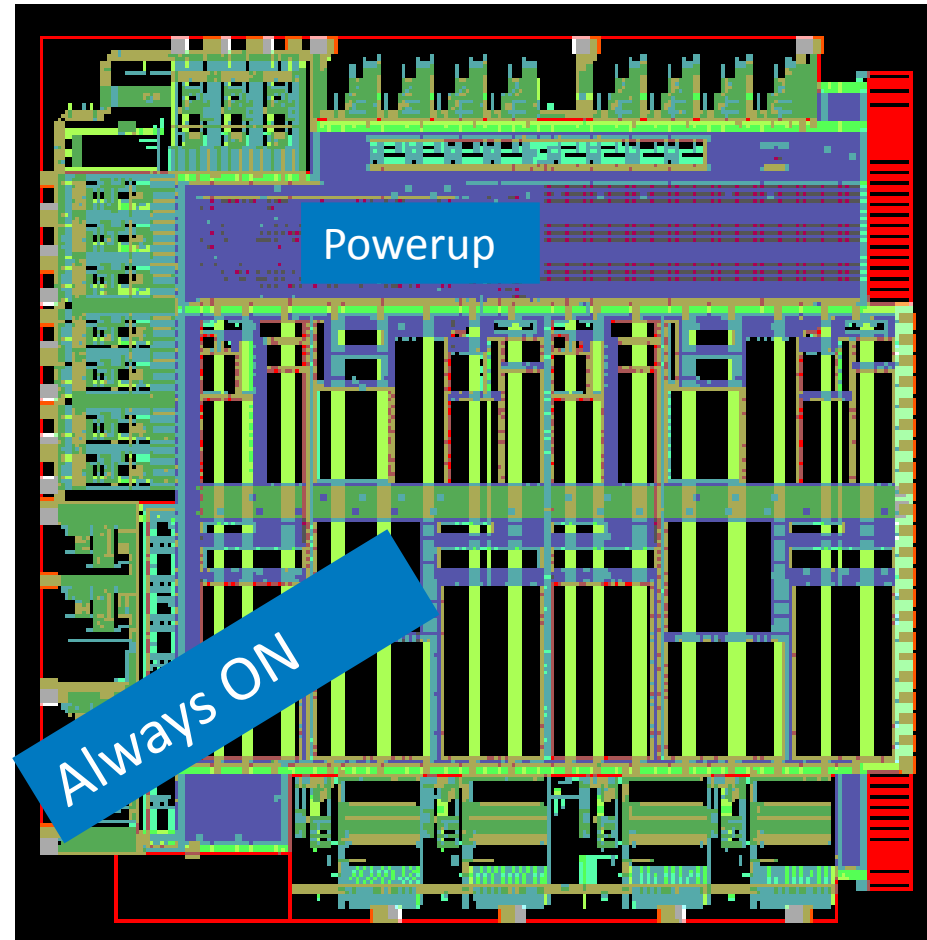
Relaxed time step

Mixed Mode Analysis

GSC File Review

```
inst_129973/* POWERUP
```

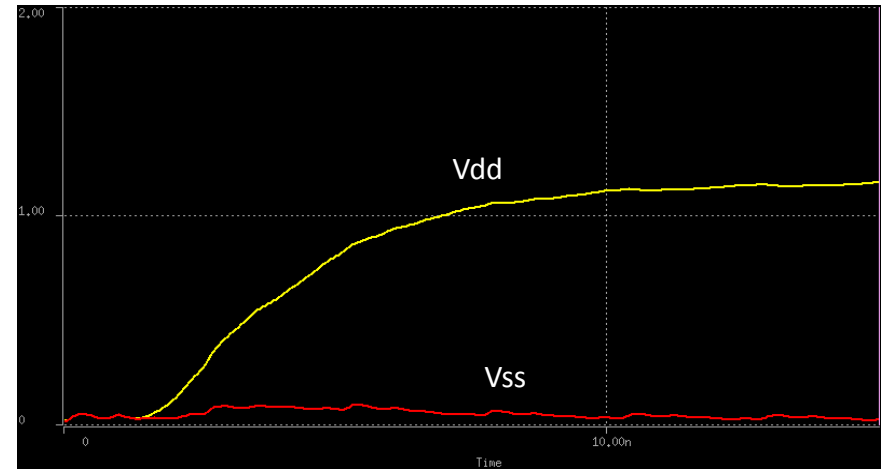
100% switches in Powerup



RedHawk – LP Lab

- **Voltage waveform for powerup instance**

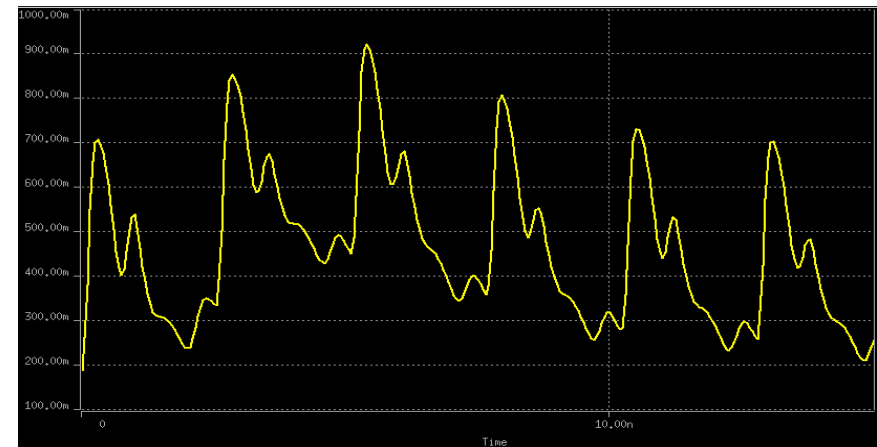
```
plot voltage -name  
inst_129973/inst_362960  
-sv
```



- **Supply current waveform**

```
plot current -net -power  
-pad -sv
```

- Shows higher battery current during powerup

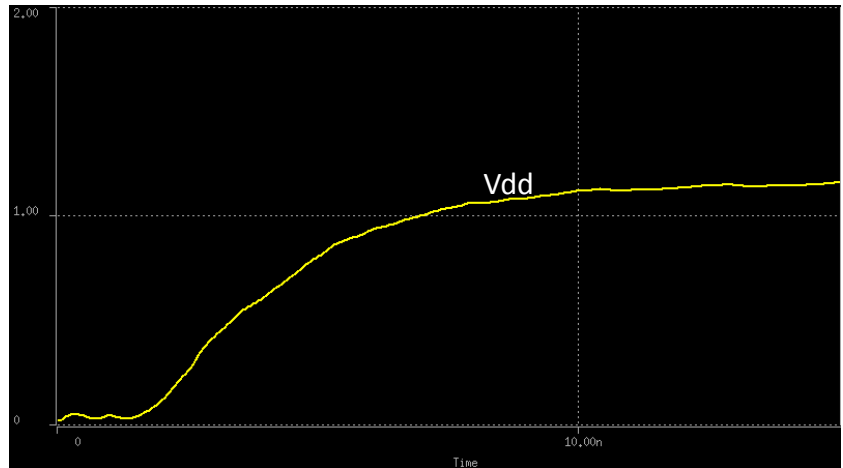


RedHawk – LP Lab (Cont'd)

- **Voltage waveform for powerup instance**

```
plot voltage -name  
inst_129973/inst_362960 -sv
```

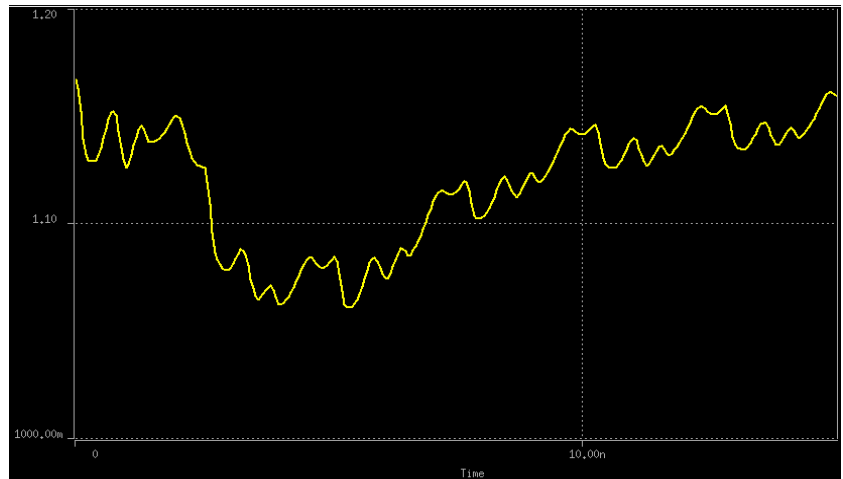
- Rampup between 3ns and 5ns



- **Voltage w/f for Always ON instance**

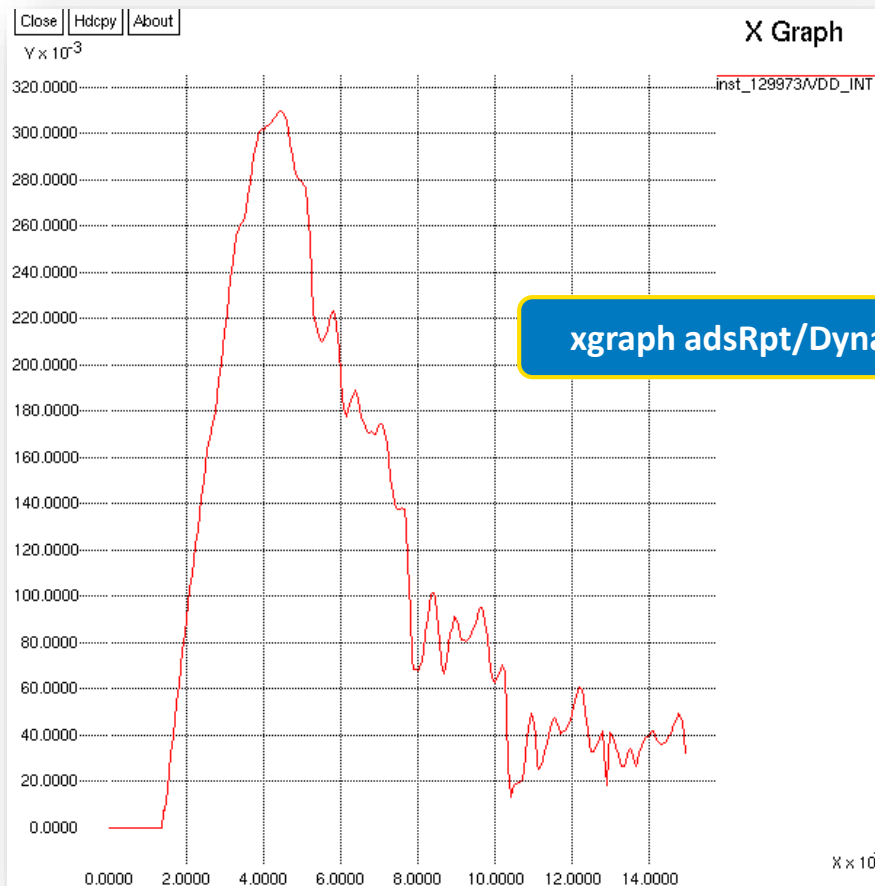
```
plot voltage -name  
inst_130017 -sv
```

- Shows higher drop during Powerup



RedHawk – LP Lab (Cont'd)

- **Voltage Domain Current W/f**
 - Represents the total current through the switches

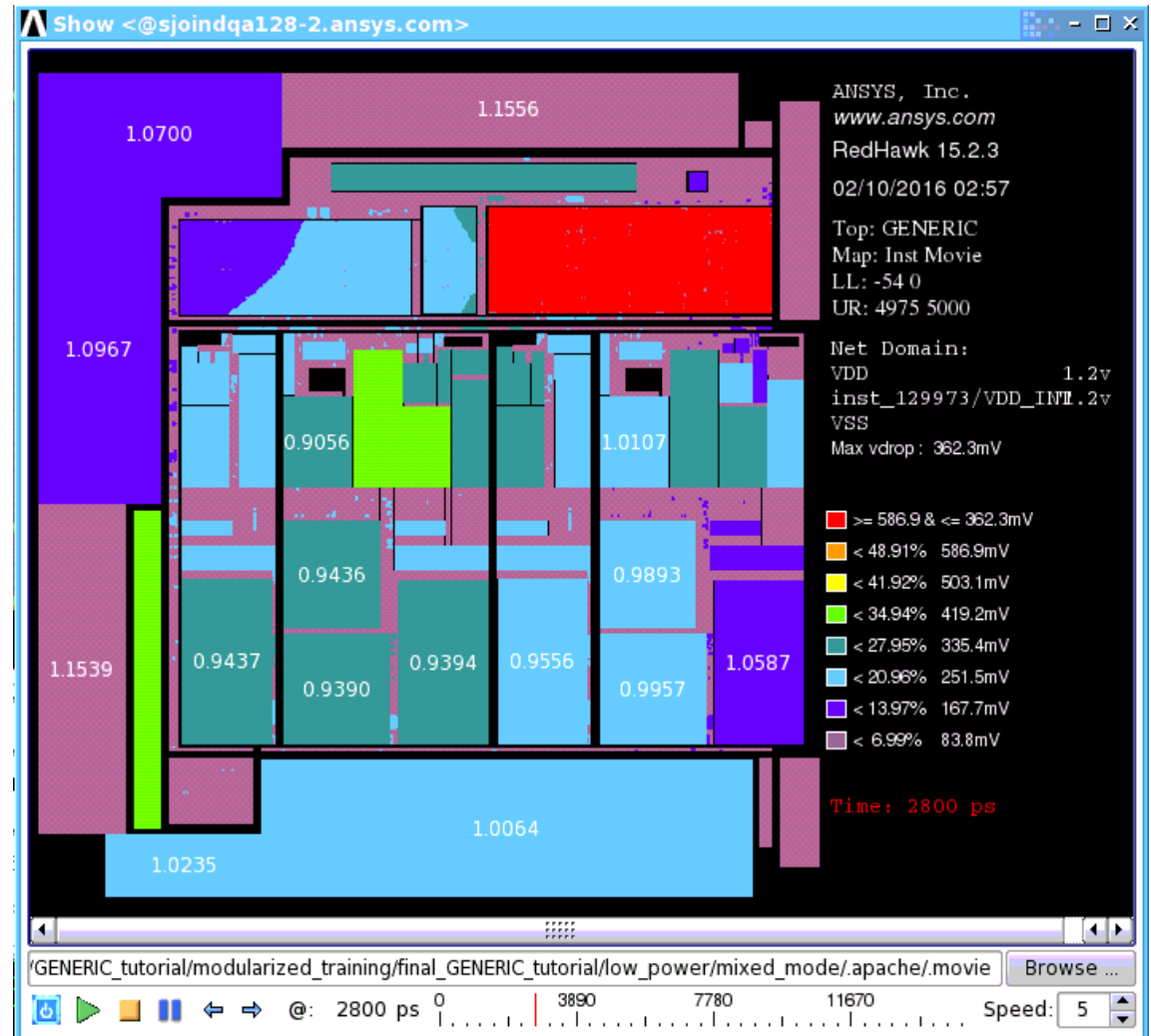


xgraph adsRpt/Dynamic/virtual_domain_total_i.rpt

RedHawk – LP Lab (Cont'd)

Rampup Movie

- Results -> Movie -> Make



Low Power Support in Explorer

- **Analysis Summary for Low Power**
- **Data integrity Checks for LP analysis**
- **Signoff Analysis Support**
 - Peak current
 - Wake-up time
 - Differential voltage check
 - Noise coupling check
 - Switch Id-sat check
 - Off state Leakage and Voltage check
- **Rampup switch turn ON Movie Generation**

View the slides in slide-show mode(shift+F5)

Low Power Summary

RedHawk™-Explorer

Summary Data Integrity Design Weakness Hotspots Performance

Back Forward 3D-IC FAQ

Design Summary

Results Summary

- Power Summary
- Static Voltage Drop Summary
- Dynamic Voltage Drop Summary
- Power EM Summary
- Signal EM Summary
- PathFinder Summary
- Low Power Check Summary
- CPM Summary

Run Details

RHE Summary » Low Power Summary

Low Power Analysis Summary

Summary Waveforms

Parameter	Value
Peak Battery Current	948.284 mA
Worst Rampup time	9.05 ns
Max Differential Voltage	82.38 mV
Worst Switch Current	3.56684 mA
Max Noise Voltage	199.39 mV

Switch Turn ON Movie

ALP Related Data Integrity Check

RedHawk™-Explorer

ANSYS

SummaryData IntegrityDesign WeaknessHotspotsPerformance

BackForward3D-ICFAQ

Library Data

APL Current Check✓

APL Cap Check✗

APL Pwcap Check✗

LIB Check✓

LEF Check✓

CMM Check

Gds2def Check✗

Design Data

DEF Check✗

SPEF Check✓

IPF Check

STA Check✓

VCD Check

RedHawk Data

GSR Tol Check✗

Package Settings✓

PAD_FILES✓

GSC FILE

Log Summary

ERROR Check✗

WARNING Check✗

Data Integrity > Apl Pwcap Check

Apl Pwcap Check

● Checks whether the APL Pwcap coverage in the design is at least 95%

✗ Number of instances with APL pwcap coverage: 300978/475418 (63.31 %)

✗ Number of cells with APL pwcap coverage: 196/237 (82.0 %)

● Apl Pwcap Area Coverage : 4.56224e+06/20853700.0 (21.87 %)

Celltype Wise CoverageCross Probing

Celltype Wise Apl Pwcap Coverage

CellType	% Coverage/Total
Memory	- % (-/-)
Combinational	99.86 % (57836/57917)
Sequential	94.38 % (15975/16926)
Clock	100.00 % (6060/6060)
Others	- % (-/-)
FILLER/DECAP	0.00 % (0/173410)
Decap	100.00 % (221105/221105)
Total	63.31 % (300978/475418)

Apl Pwcap Coverage Violations Displayed

Highlighting Regions with APL Pwcap coverage less than 95 % in GUI

36

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ALP Hotspots Section

Summary

Data Integrity

Design Weakness

Hotspots

Performance

Back

Forward

3D-IC

FAQ

Dynamic Voltage Drop Check

Static IR Check

Power EM Check

Signal EM Check

RMS

AVG

Peak

Max Cap Check

Memory Check

DVD Check

Static IR Check

Power EM Check

Low Power Check

Voltage & Current w/f

Differential Voltage Check

Noise Coupling Check

Switch Id-sat Check

Switch Off State Check

CPM

GSR/Tcl Settings

Average Current Check

FFT Check

HotSpot Summary

Analysis Summary

Parameter	Summary Description
☐ Dynamic Voltage Drop Check	Check Not Performed
☒ Static IR CHECK	Check Not Performed
☒ Power EM CHECK	Check Not Performed
☐ Low Power CHECK	Worst Differential voltage :82.38 mV

Move the mouse pointer above the table rows to highlight the corresponding Hotspots

Violation Areas

Peak Rampup Current & Rampup Time Checks

Hot Spot Summary » Voltage & Current w/f

Voltage & Current w/f

Rampup Current Check

Rampup Time Check

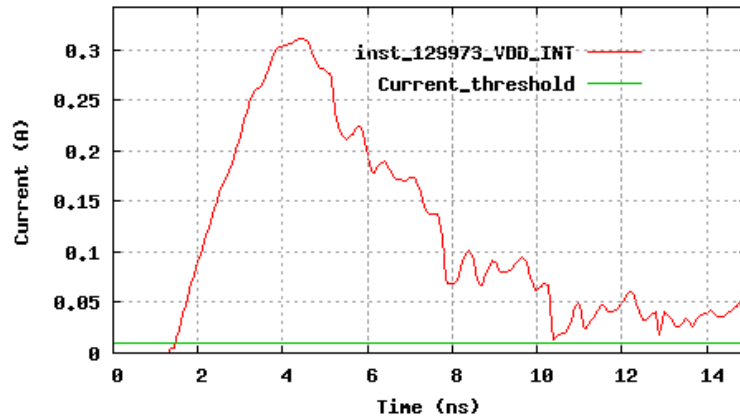
Battery Current Check

Demand Current Check

- Checks whether rampup current exceeds the threshold 10 mA
- Peak rampup current = 310.9435 mA

❌ Peak Rampup Current check fails

Rampup Current Waveform



Domain-wise Information

Internal Domain	Peak Current	Status
inst_129973_VDD_INT	310.9435 mA	❌

Hot Spot Summary » Voltage & Current w/f

Voltage & Current w/f

Rampup Current Check

Rampup Time Check

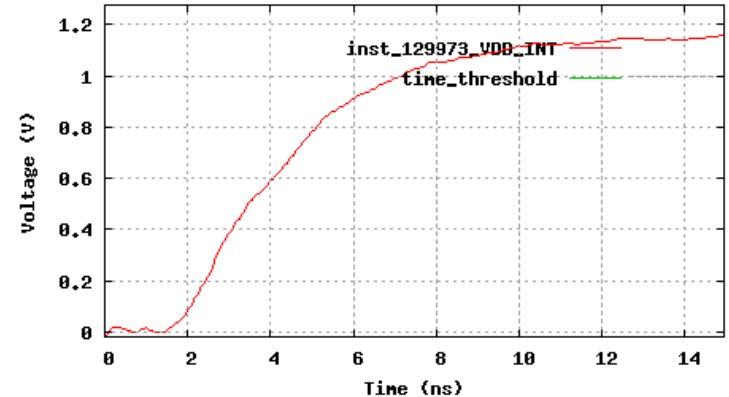
Battery Current Check

Demand Current Check

- Shows the worst ramping up/down switch
- Threshold for ramping up/down time = 150 ns
- Worst ramping up/down domain = inst_129973/VDD_INT
- Time taken to ramp up/down = 9.05 ns

✅ Rampup Time check passes

Worst ramping up/down waveform



Domain-wise Information

Internal Domain	On State Voltage	Rampup Time	Status
inst_129973/VDD_INT	1.158616 V	9.05 ns	✅

Battery Current & Demand Current Checks

Hot Spot Summary » Voltage & Current w/f

Voltage & Current w/f

Rampup Current Check

Rampup Time Check

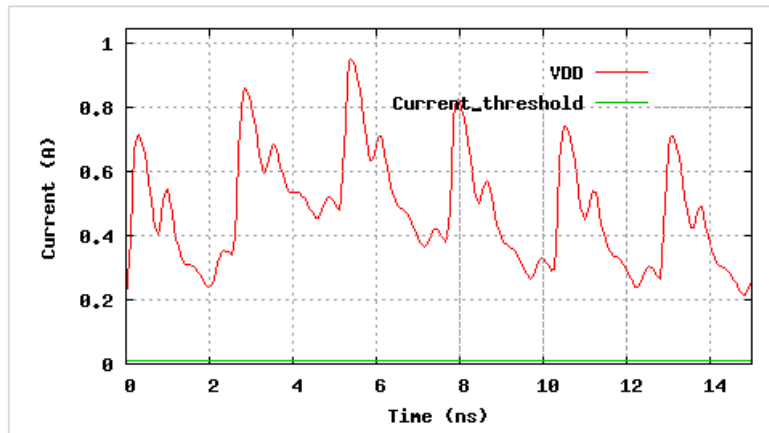
Battery Current Check

Demand Current Check

- Checks whether battery current exceeds the threshold 10 mA
- Peak battery current = 948.284 mA

❌ Peak Battery Current check fails

Battery Current Waveform



Domain-wise Information

Domain	Peak Current	Status
VDD	948.284 mA	❌

Hot Spot Summary » Voltage & Current w/f

Voltage & Current w/f

Rampup Current Check

Rampup Time Check

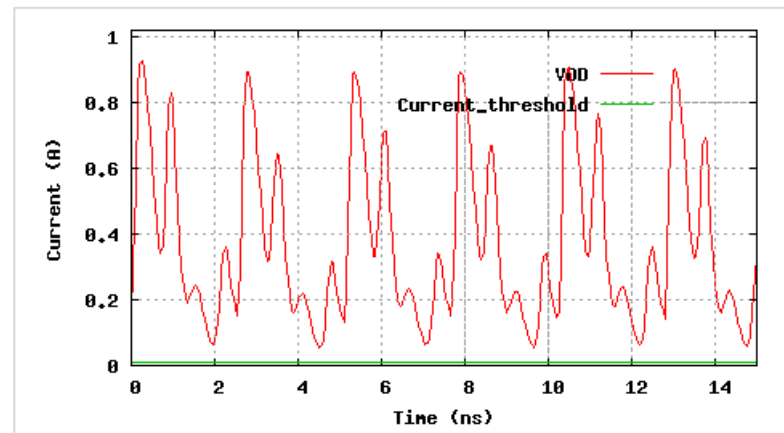
Battery Current Check

Demand Current Check

- Checks whether demand current exceeds the threshold 10 mA
- Peak demand current = 925.988 mA

❌ Peak Demand Current check fails

Demand Current Waveform



Domain-wise Information

Domain	Peak Current	Status
VDD	925.988 mA	❌

What is Differential Voltage Check ?

Hot Spot Summary » Differential Voltage Check

Differential Voltage Check ?

Differential Voltage Check:

Another challenge with power gating is the high crowbar current that can be caused due to differential voltage between driver-receiver pairs. For all driver-receiver instance pairs in the design, if the driver instance is ramping up much later than the receiver instance, Out put of the driver instance will be rising very slowly while the receiver is already in the fully ON state. This will create a high short-circuit current across the design. This scenario is illustrated in the following diagram..

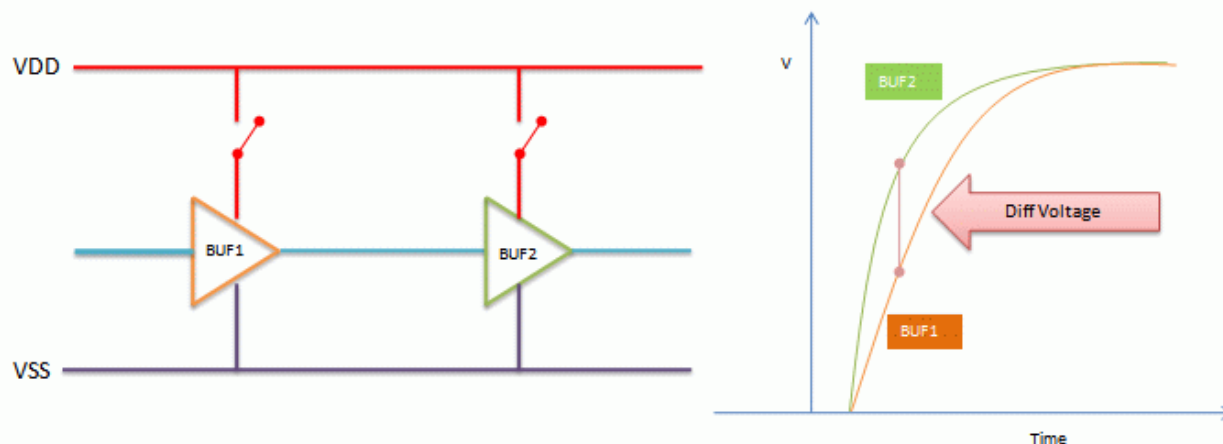


Fig 1 :
Due to high differential voltage, BUF2 may stay in ON-state while BUF1 is ramping up.
This will cause high crow-bar current through BUF2.

If this situation happens with many driver-receiver pairs in the design, it will create very high current across the chip causing damage. So, it is important to keep the differential voltage between the driver-receiver pairs within a limit.

In this check, Explorer finds all driver-receiver instance pairs in the design. Then it looks at the corresponding switch instances sitting close to it. It analyzes the voltage at the internal pin of these switches and computes the differential voltage between them. If the max differential voltage exceeds 200mv, it is reported as a violation. RHE, by default displays the top 3 violators in the report. Using the constraint editor, we can control the differential voltage threshold and number of violations to display in the report.

Differential Voltage Analysis

Hot Spot Summary » Differential Voltage Check

Differential Voltage Check

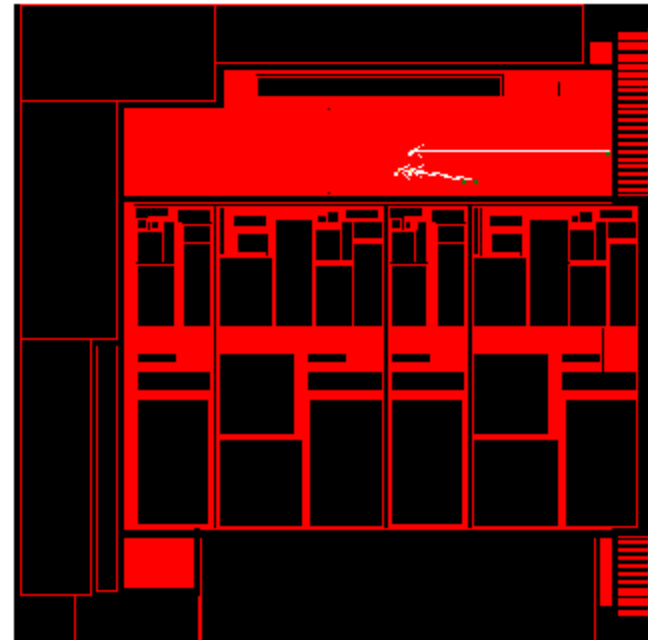
- Rampup Worst Switch Pair Exploration
- Highlighting Top 3 Worst Switches in the design
- Differential Voltage Threshold used for PASS/FAIL : 200 mV
- ✓ Number of failed pairs : 0/1511 (0.0)%

Pass/Fail Criteria Check

Top Violations

Hot Pairs	Differential Voltage	Status
 Hot Pair:1	82.38 mV	✓
 Hot Pair:2	55.53 mV	✓
 Hot Pair:3	52.784 mV	✓

List of violations



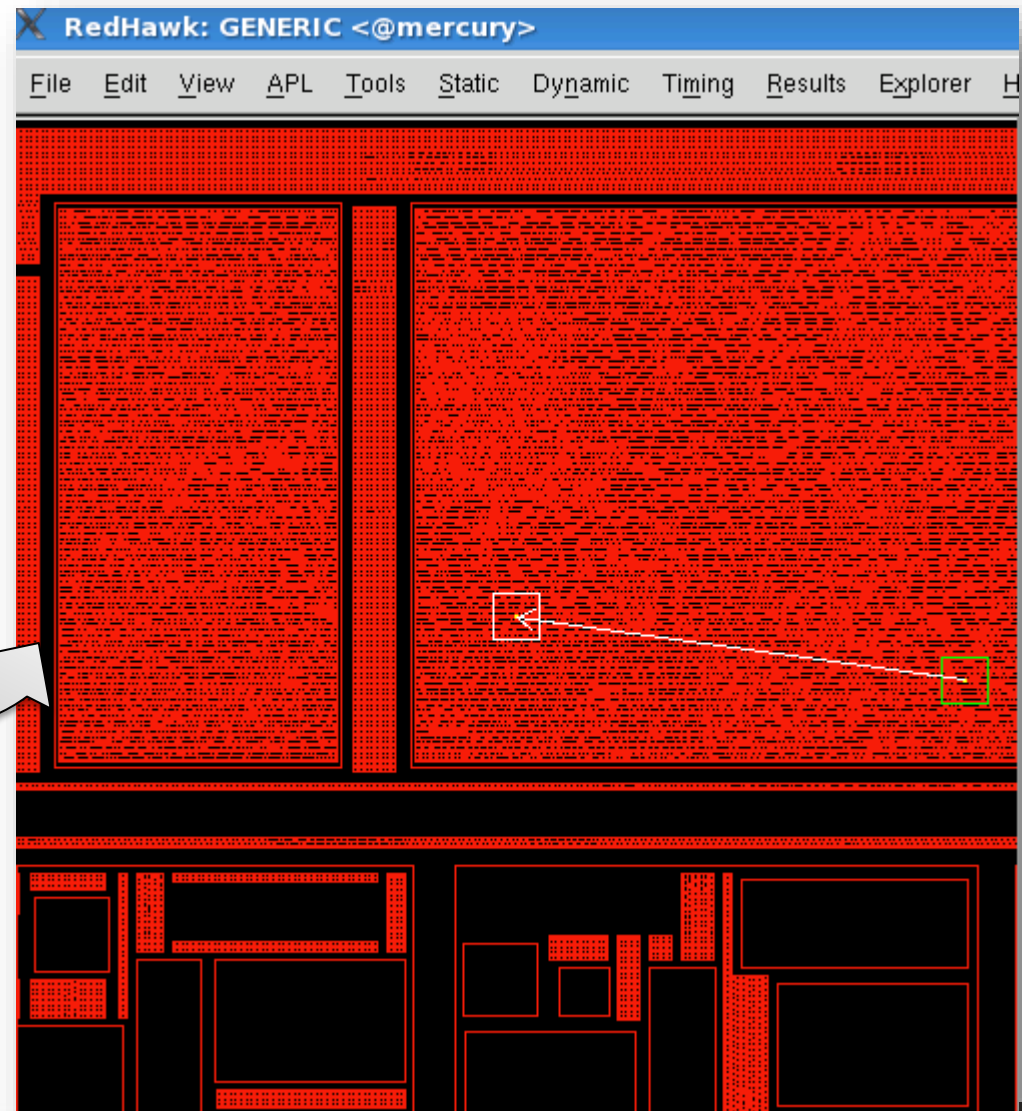
Violations highlighted in
GUI

Cross-Probing Violations

Top Violations

Hot Pairs	Differential Voltage	Status
 Hot Pair:1	82.38 mV	✓
 Hot Pair:2	55.53 mV	✓
Hot Pair:3	52.784 mV	✓

Driver and receiver instances are highlighted using different colors . Driver uses Green color and Receiver uses White color. You can choose the color using Cross-probing options.



Adjusting Differential Voltage Thresholds

User can control the diff voltage limit and number of violations to display

Differential Voltage Drop Histogram

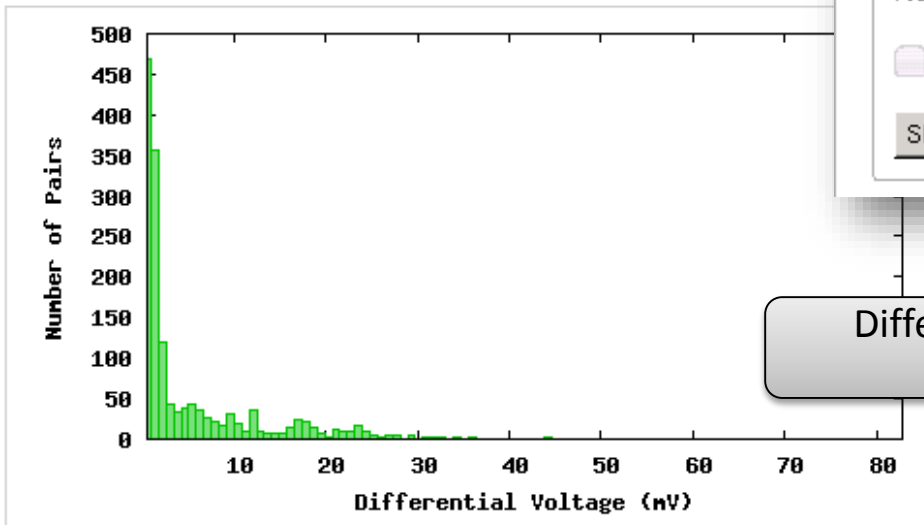


Diagram illustrating the 'RH Crossprobing Options' dialog box, which allows users to adjust differential voltage thresholds and the number of violations displayed.

The dialog box has two tabs: 'Histogram' and 'Cross Probing'. The 'Cross Probing' tab is selected.

RH Crossprobing Options

- Highlight Color:** A color selection palette with a checkmark next to the first color (red).
- Differential Voltage Limit (mV):** A slider control set to 200 mV.
- Number of Violations:** A slider control set to 3.
- Show in RedHawk:** A button to toggle the display of violations in the RedHawk interface.

Differential voltage histogram

Differential Voltage Violation Details

Hot Spot Summary » Differential Voltage Check » Hot pair 1

Hot Pair 1 ?

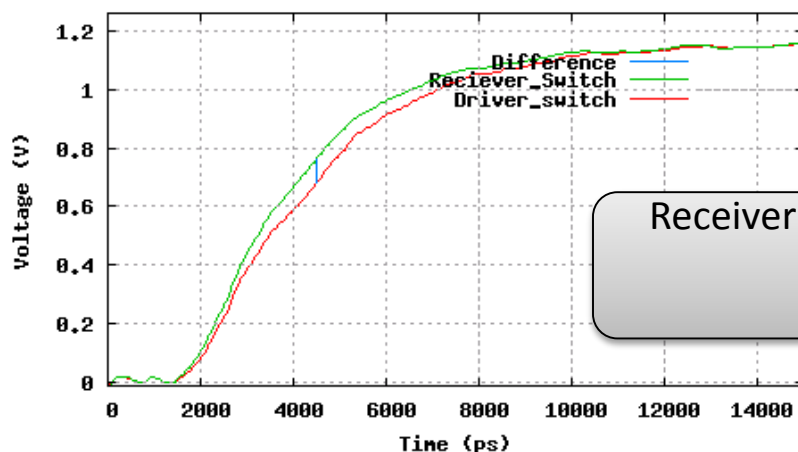
- Driver instance: inst_129973/inst_377621
- Driver Switch : inst_129973/switch_inst_R9_C52
- Driver Switch Turn-On time: 22900 22900
- Driver Switch Net Pairs : VDD:inst_129973/VDD_INT
- Receiver : inst_129973/inst_355482
- Receiver Switch: inst_129973/switch_inst_R9_C7
- Receiver Switch Turn-On time: 3775 3775
- Receiver Switch Net Pair : VDD:inst_129973/VDD_INT
- Maximum Differential voltage : 82.38 mV
- Signal Net : inst_129973/net_93421

Switch Voltage Comparison

Zoomed View

Instance Map

Switch Voltage Comparisons



Receiver Turning ON much before
Driver

Receiver will be seeing the gate voltage slowly ramping
up.
Creates high short circuit current

Differential Voltage Violation Report

Hot Spots Summary » Differential Voltage Check » Related Reports

Differential Voltage Report ?

Related Output Reports

Report	Line Count	Definition
adsRHE/adsHSE/reports/related_report.rpt	1511	Provides the list of driver receiver pairs and their differential voltage

HotSpot Summary » Differential Voltage Check » Related Reports » [adsRHE/adsHSE/reports/related_report.rpt](#)

Differential_Voltage(V) Driver_Instance (Driver switch net pairs) Receiver_Instance (Receiver switch net pairs)

0.08238 inst_129973/inst_377621 VDD:inst_129973/VDD_INT inst_129973/inst_355482 VDD:inst_129973/VDD_INT
0.05553 inst_129973/inst_507804 VDD:inst_129973/VDD_INT inst_129973/inst_375879 VDD:inst_129973/VDD_INT
0.052784 inst_129973/inst_356424 VDD:inst_129973/VDD_INT inst_129973/inst_360196 VDD:inst_129973/VDD_INT
0.050801 inst_129973/inst_507804 VDD:inst_129973/VDD_INT inst_129973/inst_375865 VDD:inst_129973/VDD_INT
0.050038 inst_129973/inst_507804 VDD:inst_129973/VDD_INT inst_129973/inst_375867 VDD:inst_129973/VDD_INT
0.04714 inst_129973/inst_359339 VDD:inst_129973/VDD_INT inst_129973/inst_360327 VDD:inst_129973/VDD_INT
0.044546 inst_129973/inst_359434 VDD:inst_129973/VDD_INT inst_129973/inst_360119 VDD:inst_129973/VDD_INT
0.043784 inst_129973/inst_375929 VDD:inst_129973/VDD_INT inst_129973/inst_375907 VDD:inst_129973/VDD_INT
0.037224 inst_129973/inst_366115 VDD:inst_129973/VDD_INT inst_129973/inst_370731 VDD:inst_129973/VDD_INT
0.036613 inst_129973/inst_366119 VDD:inst_129973/VDD_INT inst_129973/inst_366118 VDD:inst_129973/VDD_INT

Noise Coupling Check

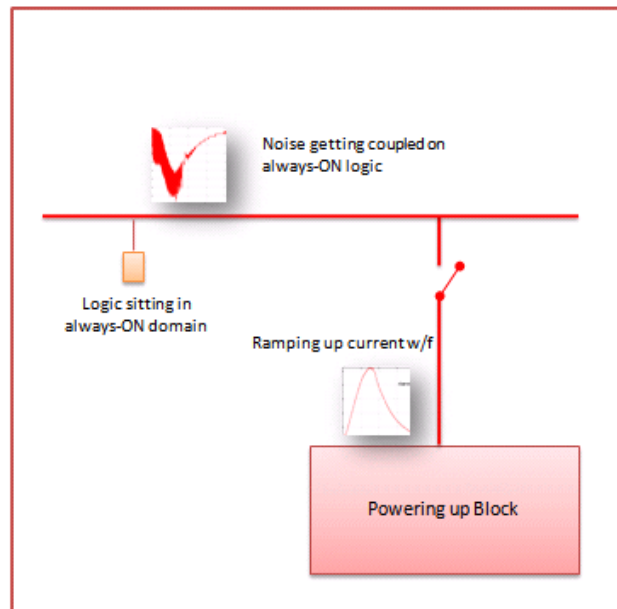
Hot Spot Summary » Noise Coupling Check

Noise Coupling Check



Noise Coupling Check

Another challenge with power gated design is the P/G noise coupling happening in the always-ON domain from the powering up block. When a block is getting powered up, it causes a high surge current in the external domain. There could be some always ON logic connected to the external domain which is supposed to be functioning all the time. Due to the high surge current flowing through the external domain, this logic will see high voltage drop and this will affect its functionality/timing. This is illustrated using the following diagram.



Noise coupling violations can be fixed by slowing down the power-up operation. This will reduce the surge current which will help reducing the coupling. But, this will have some adverse impact on the differential voltage violations.

Noise Coupling Analysis : Summary

Hot Spot Summary » Noise Coupling Check

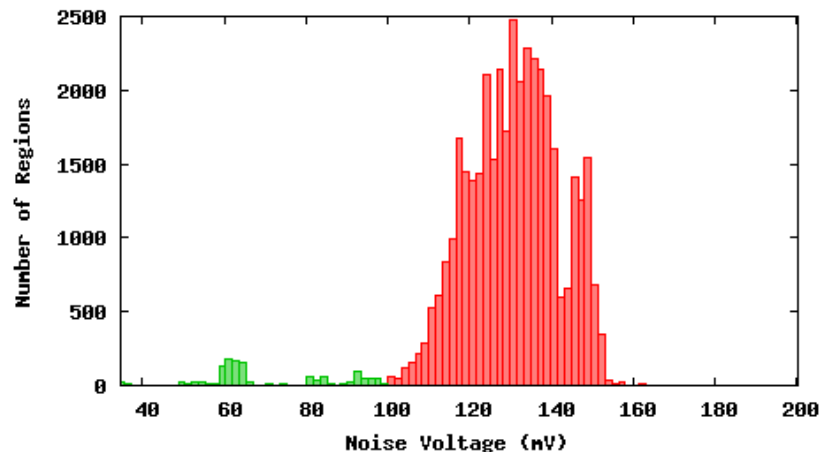
Noise Coupling Check ?

- Highlighting top 3 noise coupling hotspots in the design
- Noise coupling threshold for pass/fail 100 mV
- ⊗ Number of failing regions : 173/262 (66.03 %)
- ⊗ Number of failing instances : 38641/39900 (96.84 %)

Top Violations

Region	Drop (mV)	Failed Instances	Status
Hot Spot:1	199.39 mV	1/1 (100.0 %)	✗
Hot Spot:2	199.237 mV	1/1 (100.0 %)	✗
Hot Spot:3	172.693 mV	1/1 (100.0 %)	✗

Noise Voltage Drop Distribution Histogram



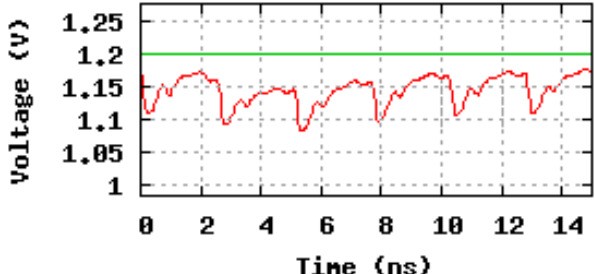
Noise Coupling Analysis Details

Hot Spot Summary » Noise Coupling Check » Hot spot 1

Noise Coupling Check Hot Spot 1 ?

- Hot Spot Region :2304.635000 2482.545000 2592.455000 2963.244000
- Instance Threshold used for PASS/FAIL : 100 mV
- Instance Name : inst_129422/inst_7778

❌ Number of failing instances 1/1 (100.0 %)

Hot Instance	Drop (mV)	Status	DVD Waveform
 Hot Instance:1	199.39	❌	DVD waveform of pin VDD 

Switch Id-sat Check

Hot Spot Summary » Switch Id-sat Check

Switch Id-sat Check

- Checks whether the current through the switch is exceeding Saturation current(Id-sat)
- Highlighting top 3 hot switches in the design
- Saturation Current Limit : Taken From Switch Model
- ✓ Number of failing switches : 0/884 (0.0 %)

Top Violations

Rank	Current (mA)	Status
 Hot Switch:1	3.56684	✓
 Hot S	3.55098	✓
 Hot	3.53147	✓

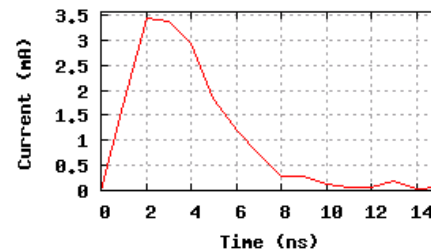


Hot Spot Summary » Switch Id-sat Check » Hot Switch 1

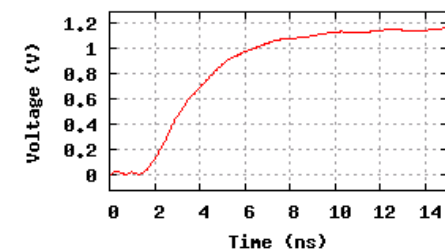
Switch Id-sat Check Hot Switch 1

- Switch Instance Name : inst_129973/switch_inst_R1_C1
- Switch Master Name : switch
- Current Through The Switch (mA) : 3.56684
- EXT:INT Net pair : VDD:inst_129973/VDD_INT
- Switch Turn On Time (pS) : 1025 1025

Switch Current Waveform



Switch Internal Pin Voltage Waveform



Highlights switch instances in the design which are driven to their saturation region due to excessive current flow during powerup

Switch Off State Check

Hot Spot Summary » Switch Off State Check

Switch Off State Check

Voltage Check

Current Check

- Checks whether off state power exceeds 20 % of on state leakage power

Top Violations

Internal Domain	Off State Power	On State Power	Status
inst_129973/VDD_INT	1.036928772e-09	2.0289e-04	✓

Voltage Check

Current Check

- Checks whether off state power exceeds 20 % of on state leakage power

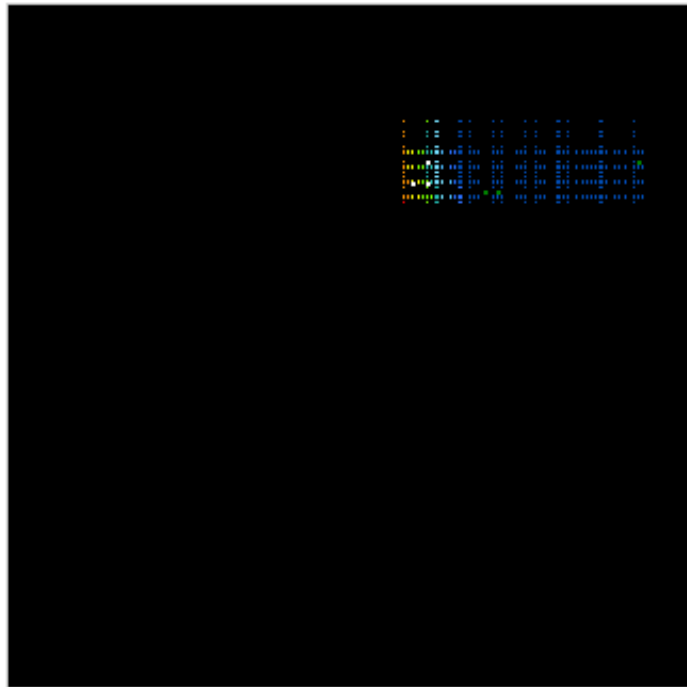
Top Violations

Internal Domain	Off State Power	On State Power	Status
inst_129973/VDD_INT	1.036928772e-09	2.0289e-04	✓

This check analyzes the effectiveness of power gating in the design in terms of reducing the leakage power

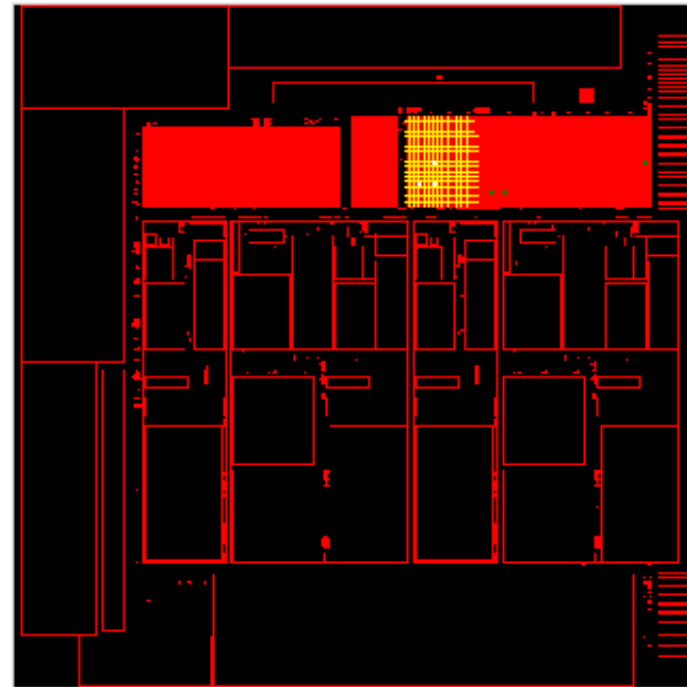
Low Power : Related Maps & Movies

Hot Spot Summary » Voltage & Current w/f » Related Maps



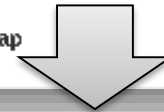
Switch Current Map

Show in RedHawk »



Switch TurnOn Movie Map

Show in RedHawk »



In the movie, you can verify that Receiver Turns on much before Driver

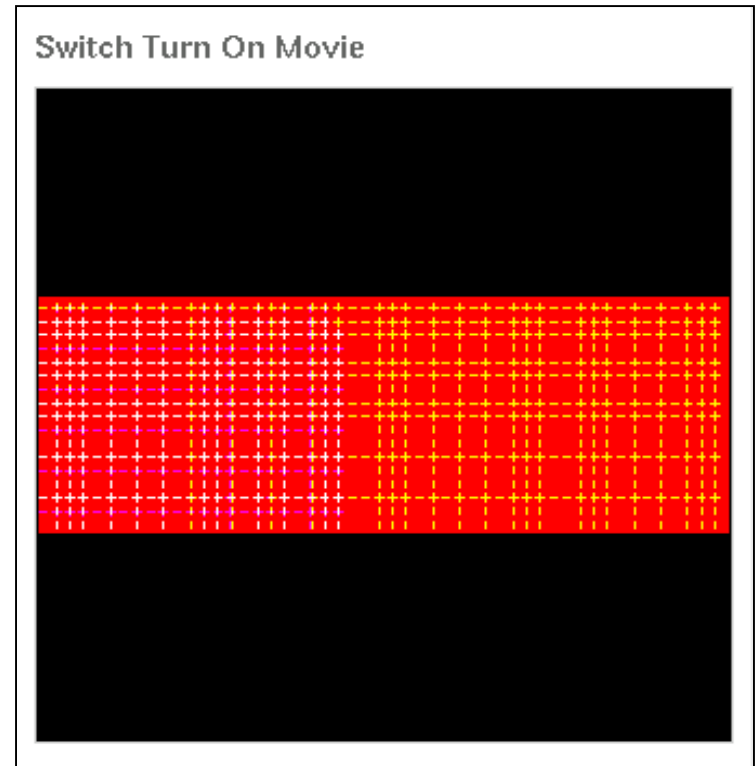
RedHawk – LP Lab Summary

- **Used RedHawk low-power analysis for**
 - Analyzing dynamic voltage drop in ON state
 - Determining the leakage current in “sleep mode”
 - Finding P/G rail voltages after “power up” and analyzing the wakeup time
 - Analyzing the effect of “rush current” on the design with the movie mode
 - Experimenting:
 - Switch turn-on time
 - Number of switches
 - Performing mixed mode Rampup analysis

ALP Movie Support For Mother-Daughter Switches

- If the design contains mother-daughter switches, RHE can now generate the switch turn on movie for both the mother switch chain and daughter switch chain.
- Mother switch chain movie will be displayed with a different color
- In order to enable this feature, turn on the following keyword in RedHawk before executing “explore design”

```
set rhe_rampup_enable_md_movie 1
```



Changing the Low Power Constraints

RedHawk Explorer Constraints Editor <@sjoindqa128-2.ansys.com>

Data Integrity Design Weakness **Hot Spots**

☒ DVD Hot Spots Check	
Maximum allowed DVD threshold(mv)	100
Number of violating buckets to display in report	3
DVD signoff parameter to be used	minWC

☒ Static IR Hot Spots Check	
Max allowed static IR threshold	100
Max allowed static IR threshold for intance(mv)	100
Number of violating buckets to display in report	3

☒ Power EM Hot Spot Check	
Maximum allowed power EM threshold(%)	100
Number of violating buckets to display in report	3

☒ Enabling Low Power Check	
Peak battery current threshold(mA)	10
Driver-receiver pair filtering distance(u)	200
Time to rampup threshold(ns)	150
Max differential voltage threshold(mv)	200
Number of differential voltage violations to report	3
Threshold for noise coupling check(mv)	100
Number of noise coupling violations	3
Fanout limit for differential voltage check	100

Bucket Size: 50

Enable All Disable All Defaults Load File Save As Apply OK Cancel

Explorer -> Generate ->
Edit Constraints

How to Get Help!

- **Apache Online Customer Support Center**
 - <http://support.apache-da.com>
 - Email: support@apache-da.com

The screenshot shows the ANSYS Apache RedHawk Support Center website. The header features the ANSYS and Apache logos, a navigation menu with links for PRODUCTS, FLOWS, SUPPORT, COMMUNITY, and COMPANY, and a search bar. The main content area is titled "RedHawk Support" and includes a sidebar with links to PowerArtist Support, RedHawk Support, Documentation, Release Notes, Known Issues, Download Instructions, Installation Instructions, Quick Start Guides, Application Notes, Training, and Resources. The main content area also displays the "RedHawk Support" title, a breadcrumb trail "Home > Support", and a description of RedHawk™ as a full-chip, dynamic power analysis and sign-off solution for high-performance SoCs. A list of technical resources is provided, including links to Documentation, Release Notes, Known Issues, Download Instructions, Installation Instructions, Quick Start Guides, Application Notes, Training, and Resources.

ANSYS Apache

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RedHawk Support

Home > Support

RedHawk Support

RedHawk™ is a full-chip, dynamic power analysis and sign-off solution for high-performance SoCs, including advanced low-power designs.

The following technical resources are available for the RedHawk products:

- [Documentation](#)
- [Release Notes](#)
- [Known Issues](#)
- [Download Instructions](#)
- [Installation Instructions](#)
- [Quick Start Guides](#)
- [Application Notes](#)
- [Training](#)
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PowerArtist Support

RedHawk Support

Documentation

Release Notes

Known Issues

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Training

Resources

Thank You!